PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Activity 3 Submission

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Topic: Linux File System Exploration, Memory Acquisition, and macOS Forensics

UE23CS342BA6

Aim:

To explore the Linux file system, analyze EXT4 metadata, perform memory acquisition using the LiME tool, collect system information using SysScout, and execute basic macOS forensic commands.

Objective:

- Understand the Linux file system structure.
- Identify mounted partitions and disk file system types.
- Analyze EXT4 file system metadata.
- Acquire a memory dump using LiME.
- Collect system information using SysScout.
- Perform basic macOS forensic commands.

Tools Required:

Host System	macOS
Virtual Machine	UTM Virtual Machine
Guest OS	Ubuntu Linux
Version Control	Git
Memory Tool	LiME - Linux Memory Extractor
Recon Tool	SysScout
Terminals	Linux terminal and MacOS Terminal

Theory Overview:

4.1 Linux File System

The Linux file system organizes data into directories and files. Modern Linux distributions commonly use the EXT4 file system, which provides journaling, improved performance, and better reliability.

4.2 Memory Forensics

Memory forensics involves capturing and analyzing the contents of volatile memory (RAM) to detect running processes, network connections, malware artifacts, and other forensic evidence.

4.3 LiME (Linux Memory Extractor)

LiME is a loadable kernel module (LKM) used to acquire a complete memory dump from a running Linux system. The dump can later be analyzed using tools such as Volatility.

4.4 SysScout
SysScout is a shell-based tool that collects system information including running processes, hardware details, and forensic artifacts from a live Linux system.

Change shell name:
PS1="shell:PES1UG23CS433:PranavHemanth:\w\n\$>"

Prerequisites — Install Required Packages:

Before starting the lab, ensure the following packages are installed on your Ubuntu VM. Open a terminal and run the commands below.

1) Update package lists

Command: sudo apt update

```
shell:PES1UG23CS433:PranavHemanth:~
$>sudo apt update
[sudo] password for seed:
Hit:1 http://ports.ubuntu.com/ubuntu-ports focal InRelease
Get:2 http://ports.ubuntu.com/ubuntu-ports focal-updates InRelease [128 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports focal-backports InRelease [128 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports focal-security InRelease [128 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports focal-updates/main arm64 DEP-11 Metad
ata [276 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports focal-updates/restricted arm64 DEP-11
 Metadata [212 B]
Get:7 http://ports.ubuntu.com/ubuntu-ports focal-updates/universe arm64 DEP-11 M
etadata [445 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports focal-updates/multiverse arm64 DEP-11
 Metadata [212 B]
Get:9 http://ports.ubuntu.com/ubuntu-ports focal-backports/main arm64 DEP-11 Met
adata [5,208 B]
Get:10 http://ports.ubuntu.com/ubuntu-ports focal-backports/restricted arm64 DEP
-11 Metadata [216 B]
Get:11 http://ports.ubuntu.com/ubuntu-ports focal-backports/universe arm64 DEP-1
1 Metadata [30.5 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports focal-backports/multiverse arm64 DEP
-11 Metadata [212 B]
Get:13 http://ports.ubuntu.com/ubuntu-ports focal-security/main arm64 DEP-11 Met
```

2) Install Git - Git is required to clone the LiME and SysScout repositories

Command: sudo apt install git -y

```
shell:PES1UG23CS433:PranavHemanth:~
$>sudo apt install git -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
git is already the newest version (1:2.25.1-1ubuntu3.14).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
shell:PES1UG23CS433:PranavHemanth:~
$>
```

Git is a version control tool that allows you to fetch, track, and manage source code repositories needed for this lab.

3) Install make and GCC - Make and GCC are needed to compile the LiME kernel module from source.

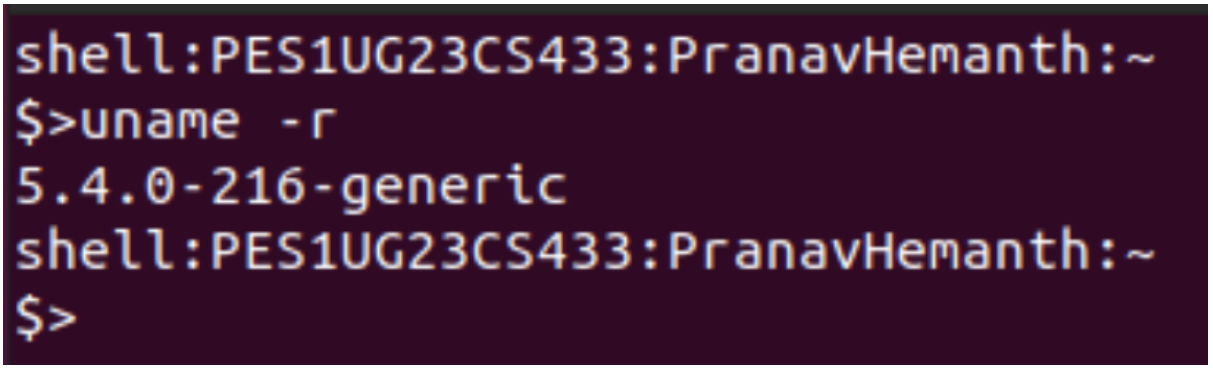
Command: sudo apt install build-essential -y

```
shell:PES1UG23CS433:PranavHemanth:~
$>sudo apt install build-essential -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-aarch64-linux-gnu binutils-common dpkg-dev fakeroot g++
  g++-9 gcc gcc-9 libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libasan5 libbinutils libc-dev-bin libc6-dev
  libcrypt-dev libctf-nobfd0 libctf0 libfakeroot libgcc-9-dev libitm1 liblsan0
  libstdc++-9-dev libtsan0 libubsan1 linux-libc-dev make manpages-dev
Suggested packages:
  binutils-doc debian-keyring gcc-9-doc gcc multilib autoconf automake libtool
  flex bison gcc-doc gcc-9-locales glibc-doc libstdc++-9-doc make-doc
The following NEW packages will be installed:
  binutils binutils-aarch64-linux-gnu binutils-common build-essential dpkg-dev
  fakeroot g++ g++-9 gcc gcc-9 libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan5 libbinutils
  libc-dev-bin libc6-dev libcrypt-dev libctf-nobfd0 libctf0 libfakeroot
  libgcc-9-dev libitm1 liblsan0 libstdc++-9-dev libtsan0 libubsan1
  linux-libc-dev make manpages-dev
0 upgraded, 30 newly installed, 0 to remove and 0 not upgraded.
Need to get 31.8 MB of archives.
After this operation, 151 MB of additional disk space will be used.
```

This installs the necessary build tools (gcc, g++, make) needed to compile kernel modules like LiME from source code.

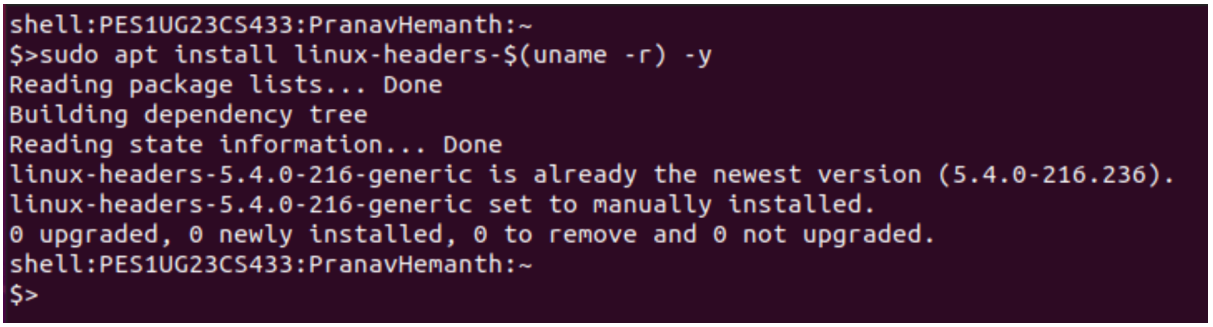
4) Install linux kernel headers - LiME compiles against your running kernel. You must install the matching kernel header files. First, check your exact kernel version:

Command: uname -r



This displays your kernel version string, such as 6.8.0-49-generic.

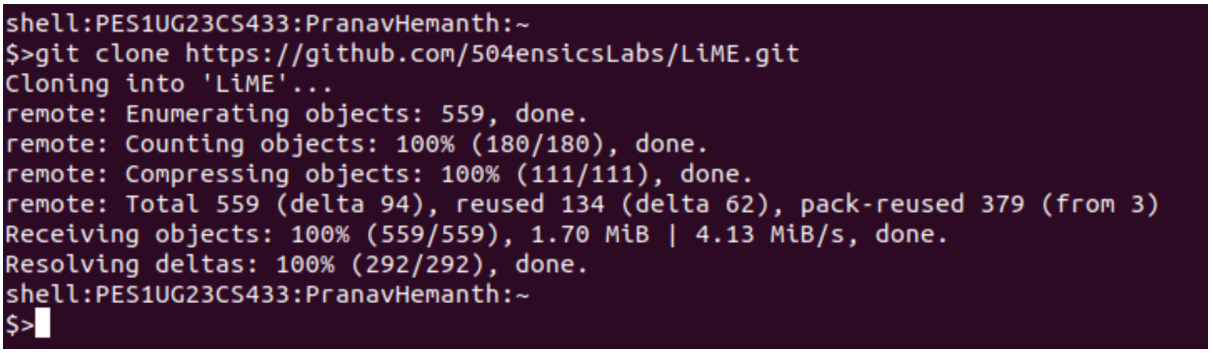
Command: sudo apt install linux-headers-$(uname -r) -y



Note: The $(uname -r) command automatically inserts your current kernel version. It also helps identify the exact .ko file generated by LiME after compilation.
Kernel headers supply the required interface definitions needed to build modules that are compatible with the running Linux kernel.

5) Clone lime

Command: git clone https://github.com/504ensicsLabs/LiME.git



6) Make the src files

Command: make

```
shell:PES1UG23CS433:PranavHemanth:~/LiME
$>ls
docs  LICENSE  LICENSES  README.md  REUSE.toml  src  test
shell:PES1UG23CS433:PranavHemanth:~/LiME
$>cd src/
shell:PES1UG23CS433:PranavHemanth:~/LiME/src
$>make
make -C /lib/modules/5.4.0-216-generic/build M="/home/seed/LiME/src" modules
make[1]: Entering directory '/usr/src/linux-headers-5.4.0-216-generic'
  CC [M]  /home/seed/LiME/src/tcp.o
  CC [M]  /home/seed/LiME/src/disk.o
  CC [M]  /home/seed/LiME/src/main.o
  CC [M]  /home/seed/LiME/src/hash.o
  CC [M]  /home/seed/LiME/src/deflate.o
  LD [M]  /home/seed/LiME/src/lime.o
  Building modules, stage 2.
  MODPOST 1 modules
  CC [M]  /home/seed/LiME/src/lime.mod.o
  LD [M]  /home/seed/LiME/src/lime.ko
make[1]: Leaving directory '/usr/src/linux-headers-5.4.0-216-generic'
strip --strip-unneeded lime.ko
mv lime.ko lime-5.4.0-216-generic.ko
shell:PES1UG23CS433:PranavHemanth:~/LiME/src
$>
```

This step builds the LiME kernel module, generating a .ko file that can be dynamically inserted into the Linux kernel.

7) How to Find the Correct .ko Filename

Command: ls LiME/src/*.ko

```
shell:PES1UG23CS433:PranavHemanth:~
$>ls LiME/src/*.ko
LiME/src/lime-5.4.0-216-generic.ko
shell:PES1UG23CS433:PranavHemanth:~
$>
```

The kernel module's filename must precisely correspond to the running kernel version to guarantee compatibility when loading it into the kernel.

8) Install sqlite 3

Command: sudo apt install sqlite3 -y

```
$>sudo apt install sqlite3 -y
[sudo] password for seed:
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  sqlite3-doc
The following NEW packages will be installed:
  sqlite3
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 790 kB of archives.
After this operation, 2,747 kB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports focal-updates/main arm64 sqlite3 arm6
4 3.31.1-4ubuntu0.7 [790 kB]
Fetched 790 kB in 4s (177 kB/s)
Selecting previously unselected package sqlite3.
(Reading database ... 157222 files and directories currently installed.)
Preparing to unpack .../sqlite3_3.31.1-4ubuntu0.7_arm64.deb ...
Unpacking sqlite3 (3.31.1-4ubuntu0.7) ...
Setting up sqlite3 (3.31.1-4ubuntu0.7) ...
Processing triggers for man-db (2.9.1-1) ...
shell:PES1UG23CS433:PranavHemanth:~
$>
```

Procedure for the lab:

Part 1: Exploring the Linux File System

9) View All Mounted Partitions

Command: df -h

```
shell:PES1UG23CS433:PranavHemanth:~
$>df -h
Filesystem                         Size  Used Avail Use% Mounted on
udev                               1.9G     0  1.9G   0% /dev
tmpfs                              392M  1.8M  391M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv  9.8G  7.2G  2.1G  78% /
tmpfs                              2.0G     0  2.0G   0% /dev/shm
tmpfs                              5.0M  4.0K  5.0M   1% /run/lock
tmpfs                              2.0G     0  2.0G   0% /sys/fs/cgroup
/dev/loop0                          62M   62M     0 100% /snap/lxd/22761
/dev/nvme0n1p2                     1.8G  112M  1.5G   7% /boot
/dev/loop2                          42M   42M     0 100% /snap/snapd/25939
/dev/loop1                          58M   58M     0 100% /snap/core20/1614
/dev/nvme0n1p1                     952M  6.4M  945M   1% /boot/efi
tmpfs                              392M   20K  392M   1% /run/user/129
tmpfs                              392M   24K  392M   1% /run/user/1000
vmhgfs-fuse                        461G  444G   18G  97% /mnt/hgfs
shell:PES1UG23CS433:PranavHemanth:~
$>
```

What to notice:

- Filesystem column: shows the device name (e.g., /dev/sda1).
- Mounted on column: shows where the partition is accessible.
- / (root partition): where EXT4 is typically used.

The df -h command displays disk usage for all mounted file systems in a human-readable format.

10) View Disk and File System Types - List all block devices with their file system types and UUIDs:

Command: lsblk -f

```
shell:PES1UG23CS433:PranavHemanth:~
$>lsblk -f
NAME FSTYPE LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINT
loop0
     squash                                                  0   100% /snap/lxd/
loop1
     squash                                                  0   100% /snap/core
loop2
     squash                                                  0   100% /snap/snap
sr0
nvme0n1
│
├─nvme0n1p1
│    vfat        1652-3150                            944.8M     1% /boot/efi
├─nvme0n1p2
│    ext4        9bcd4072-b1e9-4a93-86f0-f5fd13a89f44    1.5G     6% /boot
└─nvme0n1p3
     LVM2_m      nSaFoy-WIJc-ESLG-dkee-wBMY-CF3w-yfgZjc
  └─ubuntu--vg-ubuntu--lv
     ext4        2f0eebf0-419f-4d99-bdd6-33a426472415      2G    74% /
shell:PES1UG23CS433:PranavHemanth:~
$>
```

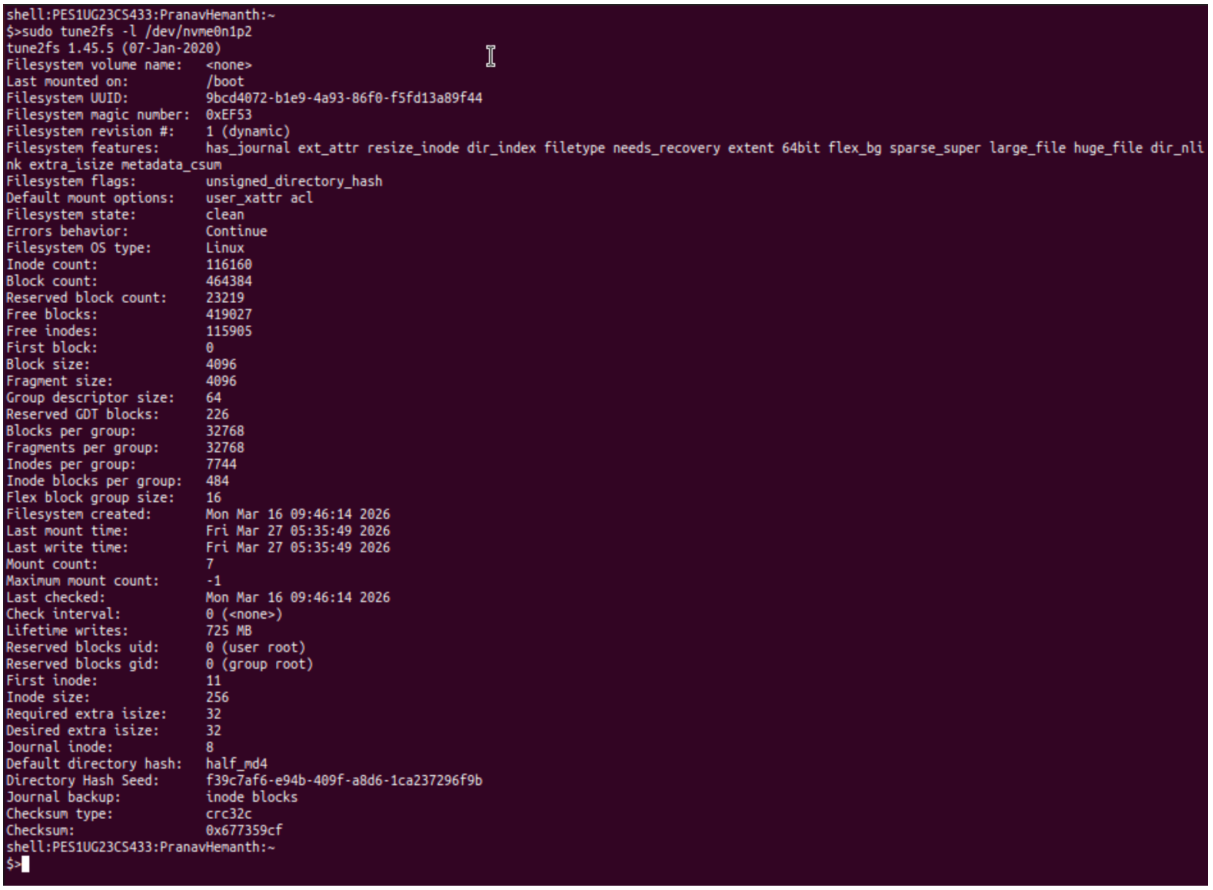
What to notice:

- FSTYPE: indicates the file system type (EXT4).
- UUID: each partition has a unique identifier.
- Forensic use: UUIDs help uniquely identify evidence partitions.

The lsblk -f command lists block devices along with their filesystem type, labels, and UUIDs, helping identify storage devices during forensic analysis.

11) Read the EXT4 Superblock (my os doesnt have EXT, so running for nvme) - Read the EXT4 superblock metadata to understand the file system structure:

Command: sudo tune2fs -l /dev/nvme0n1p2



What to notice:

- Magic number (0xEF53): confirms this is an EXT4 partition.
- Block size: data stored in 4096-byte blocks.
- Last mount time: shows when the disk was last accessed.
- Inode count: total files/directories the filesystem supports.

The tune2fs -l command retrieves filesystem metadata from the superblock, which holds key structural details of the EXT4 filesystem.

Part 2: Memory Acquisition Using LiME

12)Download LiME

Command: git clone https://github.com/504ensicsLabs/LiME.git

```
shell:PES1UG23CS433:PranavHemanth:~
$>git clone https://github.com/504ensicsLabs/LiME.git
Cloning into 'LiME'...
remote: Enumerating objects: 559, done.
remote: Counting objects: 100% (180/180), done.
remote: Compressing objects: 100% (111/111), done.
remote: Total 559 (delta 94), reused 134 (delta 62), pack-reused 379 (from 3)
Receiving objects: 100% (559/559), 1.70 MiB | 4.13 MiB/s, done.
Resolving deltas: 100% (292/292), done.
shell:PES1UG23CS433:PranavHemanth:~
$>
```

13)Compile LiME

Command:

cd LiME/src
make clean
ls
make

```
shell:PES1UG23CS433:PranavHemanth:~/LiME
$>ls
docs  LICENSE  LICENSES  README.md  REUSE.toml  src  test
shell:PES1UG23CS433:PranavHemanth:~/LiME
$>cd src/
shell:PES1UG23CS433:PranavHemanth:~/LiME/src
$>make
make -C /lib/modules/5.4.0-216-generic/build M="/home/seed/LiME/src" modules
make[1]: Entering directory '/usr/src/linux-headers-5.4.0-216-generic'
  CC [M]  /home/seed/LiME/src/tcp.o
  CC [M]  /home/seed/LiME/src/disk.o
  CC [M]  /home/seed/LiME/src/main.o
  CC [M]  /home/seed/LiME/src/hash.o
  CC [M]  /home/seed/LiME/src/deflate.o
  LD [M]  /home/seed/LiME/src/lime.o
  Building modules, stage 2.
  MODPOST 1 modules
  CC [M]  /home/seed/LiME/src/lime.mod.o
  LD [M]  /home/seed/LiME/src/lime.ko
make[1]: Leaving directory '/usr/src/linux-headers-5.4.0-216-generic'
strip --strip-unneeded lime.ko
mv lime.ko lime-5.4.0-216-generic.ko
shell:PES1UG23CS433:PranavHemanth:~/LiME/src
$>
```

14)Find Your Kernel Version (for .ko filename)

Command: uname -r

```
shell:PES1UG23CS433:PranavHemanth:~
$>uname -r
5.4.0-216-generic
shell:PES1UG23CS433:PranavHemanth:~
$>
```

Command: ls LiME/src/*.ko

```
shell:PES1UG23CS433:PranavHemanth:~
$>ls LiME/src/*.ko
LiME/src/lime-5.4.0-216-generic.ko
shell:PES1UG23CS433:PranavHemanth:~
$>
```

15)Acquire Memory Dump

Command: sudo insmod lime-5.4.0-216-generic.ko path=../Linux_Memory.mem format=raw

Command: sudo insmod lime-$(uname -r).ko path=../Linux_Memory.mem format=raw

```
shell:PES1UG23CS433:PranavHemanth:~
$>cd LiME/
shell:PES1UG23CS433:PranavHemanth:~/LiME
$>c d src/
c: command not found
shell:PES1UG23CS433:PranavHemanth:~/LiME
$>cd src/
shell:PES1UG23CS433:PranavHemanth:~/LiME/src
$>sudo insmod lime-5.4.0-216-generic.ko path=../Linux_Memory.mem format=raw
shell:PES1UG23CS433:PranavHemanth:~/LiME/src
$>
```

This loads LiME into the kernel and captures a complete memory image in raw format. By inserting the LiME module, it gains direct access to physical memory, making it possible to acquire a full memory dump for forensic analysis.

16)Verify Memory Dump

Command: ls -lh ../Linux_Memory.mem

```
shell:PES1UG23CS433:PranavHemanth:~/LiME/src
$>ls -lh ../Linux_Memory.mem
-r--r--r-- 1 root root 2.6G Mar 31 04:31 ../Linux_Memory.mem
shell:PES1UG23CS433:PranavHemanth:~/LiME/src
$>
```

The file size should correspond to the total amount of RAM installed on the system. Checking the memory dump confirms that the acquisition was successful and that the captured image is complete.

Part 3: System Information Collection Using SysScout

17)Download SysScout - Navigate to the Desktop and clone the SysScout repository:

Command:

cd ~/Desktop

git clone https://github.com/joshbrunty/SysScout.git

```
shell:PES1UG23CS433:PranavHemanth:~
$>cd ~/Desktop
shell:PES1UG23CS433:PranavHemanth:~/Desktop
$>git clone https://github.com/joshbrunty/SysScout.git
Cloning into 'SysScout'...
remote: Enumerating objects: 108, done.
remote: Total 108 (delta 0), reused 0 (delta 0), pack-reused 108 (from 1)
Receiving objects: 100% (108/108), 27.11 KiB | 274.00 KiB/s, done.
Resolving deltas: 100% (50/50), done.
shell:PES1UG23CS433:PranavHemanth:~/Desktop
$>
```

18)Run SysScout - Navigate into the SysScout directory, grant execute permission, and run:

Command:

cd SysScout

chmod +x SysScout.sh

./SysScout.sh

```
shell:PES1UG23CS433:PranavHemanth:~/Desktop
$>cd SysScout
shell:PES1UG23CS433:PranavHemanth:~/.../SysScout
$>chmod +x SysScout.sh
shell:PES1UG23CS433:PranavHemanth:~/.../SysScout
$>
```

SysScout streamlines the gathering of system and network data, making it valuable for incident response and live forensic analysis.

```
------------------------------------------


  _____                 _____                       _
 / ____|               / ____|                     | |
| (___   _   _  ___  | (___    ___   ___   _   _  | |_
 \___ \ | | | |/ __| \___ \  / __|/ _ \ | | | || __|
 ____) || |_| |\__ \ ____) || (__| (_) || |_| || |_
|_____/  \__, ||___/|_____/  \___|\___/  \__,_| \__|
          __/ |v.1.0.3
         |___/
------------------------------------------
A Network Forensics/Incident Response Tool
By: Josh Brunty: josh [dot] brunty [at] marshall [dot] edu
------------------------------------------
Current Local Machine Date & Time : Tue 31 Mar 2026 09:48:43 AM EDT
--------------------------
   Main Menu
--------------------------
1. Operating System Info
2. Time Info
3. HOST and DNS Info
4. Network Info
5. Who is Online
6. Last Logged In Users
7. Memory Information
8. Exit
Enter your choice [ 1 - 8 ]:
```

SysScout displays an interactive menu. Try the following options:

• Option 1: Collect basic system information (hostname, OS, uptime).

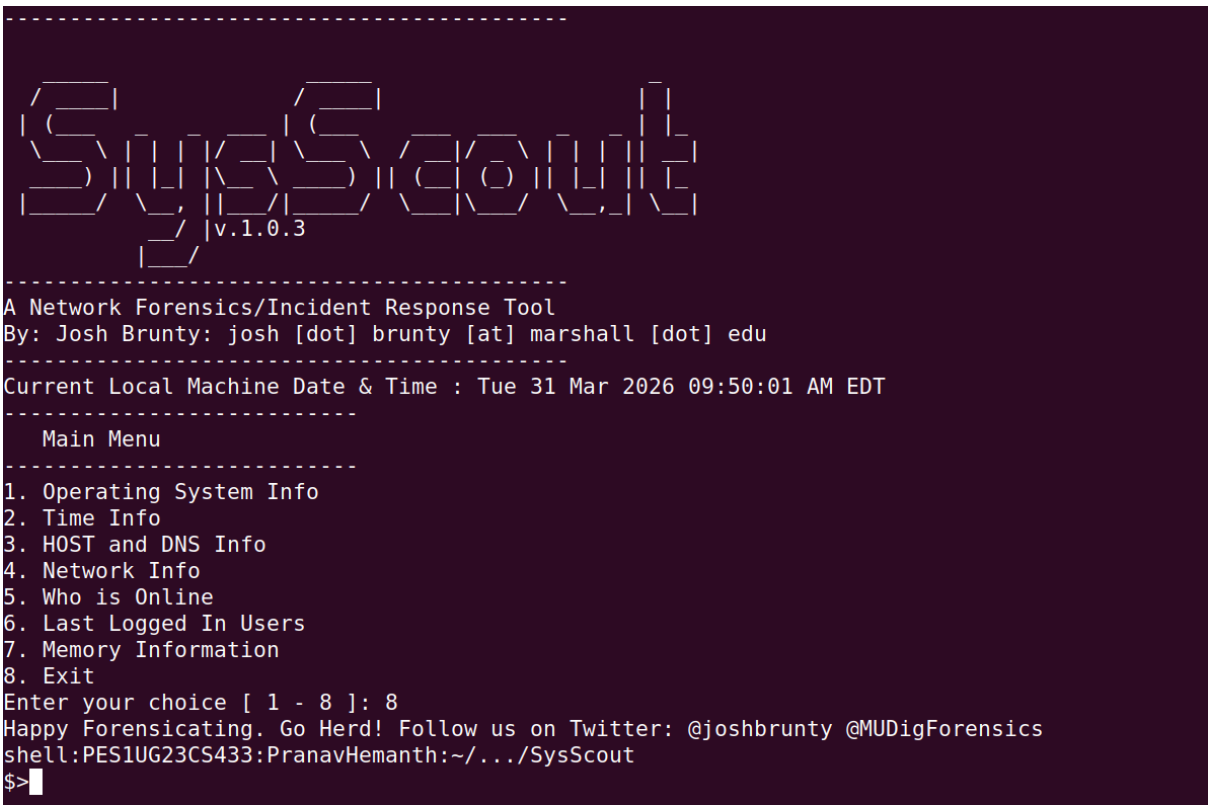
```
-----------------------------------------


  _____                 _____                    _
 / ____|               / ____|                  | |
| (___   _   _  ___  | (___    ___  ___   _   _ | |_
 \___ \ | | | |/ __| \___ \  / __|/ _ \ | | | || __|
 ____) || |_| |\__ \ ____) || (__| (_) || |_| || |_
|_____/  \__, ||___/|_____/  \___|\___/  \__,_| \__|
          __/ |v.1.0.3
         |___/
-----------------------------------------
A Network Forensics/Incident Response Tool
By: Josh Brunty: josh [dot] brunty [at] marshall [dot] edu
-----------------------------------------
Current Local Machine Date & Time : Tue 31 Mar 2026 09:48:43 AM EDT
---------------------------
   Main Menu
---------------------------
1. Operating System Info
2. Time Info
3. HOST and DNS Info
4. Network Info
5. Who is Online
6. Last Logged In Users
7. Memory Information
8. Exit
Enter your choice [ 1 - 8 ]: 1
-----------------------------------------------------------------
      Operating System Information
-----------------------------------------------------------------
Operating system : VM GNU/Linux
Operating System Version : #60-Ubuntu SMP Fri Nov 6 10:37:59 UTC 2020 x86_64
```

• Option 7: Gather extended forensic artifacts.

```
   Main Menu
---------------------------
1. Operating System Info
2. Time Info
3. HOST and DNS Info
4. Network Info
5. Who is Online
6. Last Logged In Users
7. Memory Information
8. Exit
Enter your choice [ 1 - 8 ]: 7
-----------------------------------------------------------------
     Free and used memory
-----------------------------------------------------------------
              total        used        free      shared  buff/cache   available
Mem:           1987         996          87          16         903         819
Swap:          2047          38        2009
--------------------------------
--- Virtual Memory Statistics ---
--------------------------------
procs -----------memory---------- ---swap-- -----io---- -system-- ------cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
 0  0  39852  90088  25716 899124    1   15   516  1022  154  451  1  4 95  0  0
---------------------------------------
--- Top 5 Memory Utilizing Processes ---
---------------------------------------
systemd+    1691  0.6 16.5 1283664 336616 pts/0  Ssl+ 09:00   0:18      \_ mysqld --default-authentication-plugin=mysql_native_password
seed        2171  1.3 12.1 3625424 248164 ?      Rsl  09:02   0:37  \_ /usr/bin/gnome-shell
seed        1953  0.5  2.2 249420 45888 tty2     Rl+  09:02   0:16          \_ /usr/lib/xorg/Xorg vt2 -displayfd 3 -auth /run/user/1000/gdm/X
authority -background none -noreset -keeptty -verbose 3
seed        2650  0.2  1.9 814376 39352 ?        Ssl  09:02   0:06  \_ /usr/libexec/gnome-terminal-server
root         712  0.1  1.8 795516 36824 ?        Ssl  09:00   0:05 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

• Option 8:



These options enable investigators to rapidly collect important forensic artifacts, including system information, user activity, and memory usage.

Part 4: Understanding Files and Inodes (Linux & macOS)

In Linux and macOS, everything is treated as a file. This section explores different file types and inode metadata, which are crucial in digital forensic analysis.

Device Naming Conventions: sda vs vda vs disk0

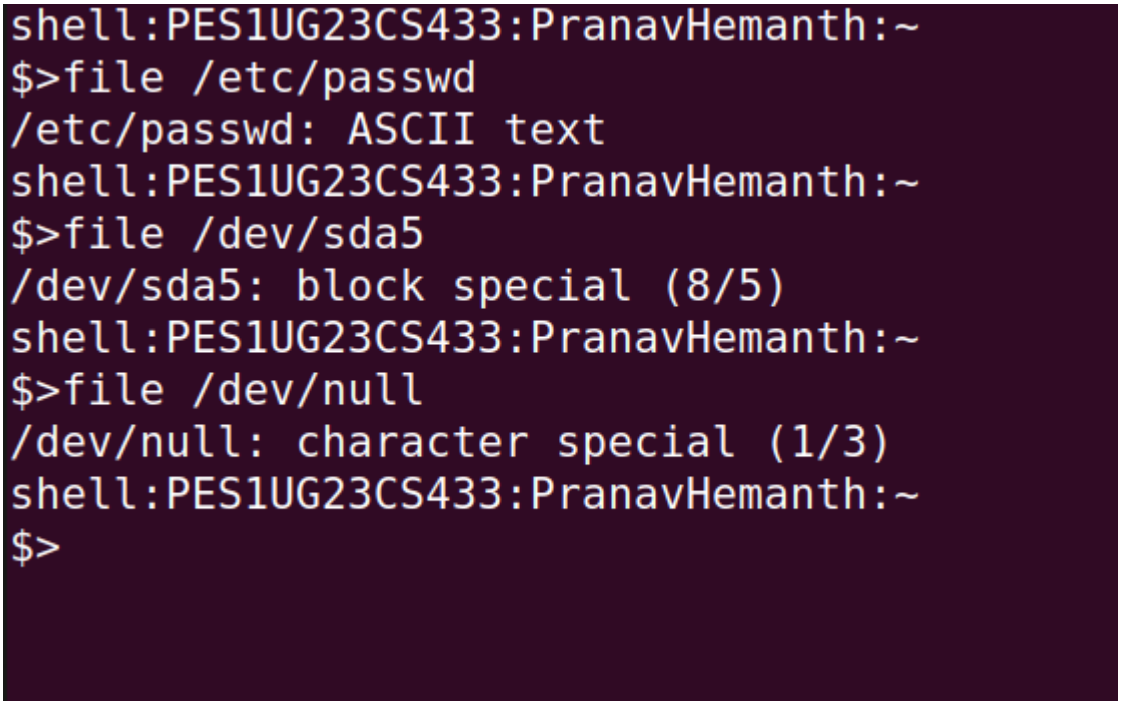
Before running commands, it is important to understand why disk device names look different across systems. The device name depends on the type of machine and OS you are using.

Device Name	Where you see it	What it means
sda, sdb, sdc...	Linux (physical or SATA disk)	SCSI/SATA disk. sda = 1st disk, sdb = 2nd disk, sda1 = 1st partition
vda, vdb...	Linux inside a Virtual Machine (UTM, KVM, QEMU)	Virtual disk. vda = 1st virtual disk, vda1 = 1st partition on it
nvme0n1...	Linux with NVMe SSD	NVMe storage. nvme0n1 = 1st drive, nvme0n1p1 = 1st partition
disk0, disk1...	macOS (all disk types)	macOS names all disks as disk0, disk1. disk0s1 = 1st partition of disk0

Note: In this lab you are using Ubuntu inside UTM (a virtual machine), so your disk will appear as vda (not sda). On a real physical Linux machine it would be sda. On macOS it is always disk0. Always check with lsblk -f or diskutil list first to confirm your exact device name before running commands.
Understanding file types and inode metadata is essential in digital forensics, as it helps track file activity, ownership, and changes over time.

19)Check File Type (linux)
Run the following commands to identify the type of different system files on Linux:
Command:

file /etc/passwd
file /dev/sda5
file /dev/null



20)Check File Type (macos)
Run the following commands to identify the type of different system files on macos:
Command:

file /etc/passwd
file /dev/disk0
file /dev/null

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %file /etc/passwd
file /dev/disk0
file /dev/null
/etc/passwd: ASCII text
/dev/disk0: block special (1/0)
/dev/null: character special (3/2)
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

What to notice:
• /etc/passwd: is a regular text file.
• /dev/sda / /dev/disk0: is a block device (represents storage disk).
• /dev/null: is a character device (discards output).
• In Linux/macOS, everything is treated as a file.

The file command identifies a file's type by analyzing its contents instead of relying on its extension.

21)Identify File Type from Listing (linux)
Command: ls -l /etc/passwd /dev/sda5 /dev/null

```
shell:PES1UG23CS433:PranavHemanth:~
$>ls -l /etc/passwd /dev/sda5 /dev/null
crw-rw-rw- 1 root root 1, 3 Mar 31 09:00 /dev/null
brw-rw---- 1 root disk 8, 5 Mar 31 09:00 /dev/sda5
-rw-r--r-- 1 root root 2886 Nov 24  2020 /etc/passwd
shell:PES1UG23CS433:PranavHemanth:~
$>
```

22)Identify File Type from Listing (macos)
Command: ls -l /etc/passwd /dev/disk0 /dev/null

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %ls -l /etc/passwd /dev/disk0 /dev/null
brw-r-----  1 root  operator  0x1000000 Mar 25 22:04 /dev/disk0
crw-rw-rw-  1 root  wheel     0x3000002 Mar 31 19:22 /dev/null
-rw-r--r--  1 root  wheel          9344 Feb 25 09:11 /etc/passwd
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

What to notice - first character indicates file type:

Symbol	Meaning
-	Regular file
d	Directory
b	Block device
c	Character device
l	Symbolic link
p	Named pipe

The first character in the file listing indicates the file type, which helps distinguish between regular files, directories, and device files.

23) Read Inode Metadata of a File (linux) - The stat command shows detailed inode metadata for a file

Command: stat /etc/passwd

```
shell:PES1UG23CS433:PranavHemanth:~
$>stat /etc/passwd
  File: /etc/passwd
  Size: 2886            Blocks: 8          IO Block: 4096   regular file
Device: 805h/2053d      Inode: 2362043     Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-03-31 00:17:21.068000388 -0400
Modify: 2020-11-24 11:12:11.442571648 -0500
Change: 2020-11-24 11:12:11.446573648 -0500
 Birth: -
shell:PES1UG23CS433:PranavHemanth:~
$>
```

24) Read Inode Metadata of a File (macos) - The stat command shows detailed inode metadata for a file

Command: stat /etc/passwd

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %stat /etc/passwd
16777232 92441811 -rw-r--r-- 1 root wheel 0 9344 "Feb 25 09:11:32 2026" "Feb 25 09:11:32 2026" "Mar 19 02:03:59 2026" "Feb 25 09:11:32 2026" 4096 8 0x20 /etc/passwd
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

The stat command displays detailed file metadata, including timestamps and inode information, which are essential for constructing forensic timelines.

What to notice:

- Inode: unique identifier of the file on the filesystem.
- Links: number of references/hard links to the file.
- Access (atime): last time the file was read.
- Modify (mtime): last time the file content was changed.
- Change (ctime): last time the file metadata was changed.
- These timestamps are very important in forensic timeline analysis.

25)Get Inode Number of a File (linux)
Command: ls -i /etc/passwd

```
shell:PES1UG23CS433:PranavHemanth:~
$>ls -i /etc/passwd
2362043 /etc/passwd
shell:PES1UG23CS433:PranavHemanth:~
$>
```

26)Get Inode Number of a File (macos)
Command: ls -i /etc/passwd

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %ls -i /etc/passwd
92441811 /etc/passwd
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

Inode numbers serve as unique identifiers for files within a filesystem and are used by the operating system to manage how data is stored.

Part 5: macOS Forensic Commands

27)Display Hardware Information
Command: system_profiler SPHardwareDataType

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %system_profiler SPHardwareDataType
2026-03-31 18:48:23.057 system_profiler[40493:6043826] hw.cpufamily: 0xfa33415e
Hardware:

    Hardware Overview:

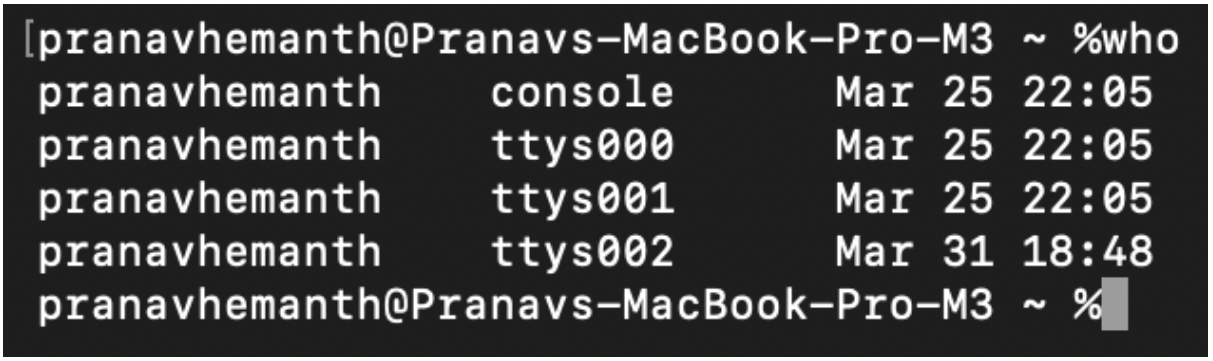
      Model Name: MacBook Pro
      Model Identifier: Mac15,3
      Model Number: MTL73LL/A
      Chip: Apple M3
      Total Number of Cores: 8 (4 Performance and 4 Efficiency)
      Memory: 8 GB
      System Firmware Version: 13822.81.10
      OS Loader Version: 13822.81.10
      Serial Number (system): LGY27M372L
      Hardware UUID: 6097649B-E786-571C-A638-D736D0519857
      Provisioning UDID: 00008122-001178423A40001C
      Activation Lock Status: Enabled

pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

This command gathers comprehensive hardware and system configuration details, helping identify the system being examined.

28)Show Logged-in Users

Command: who



The who command lists users who are currently logged in, allowing you to identify active sessions on the system.

29)Display Running Processes

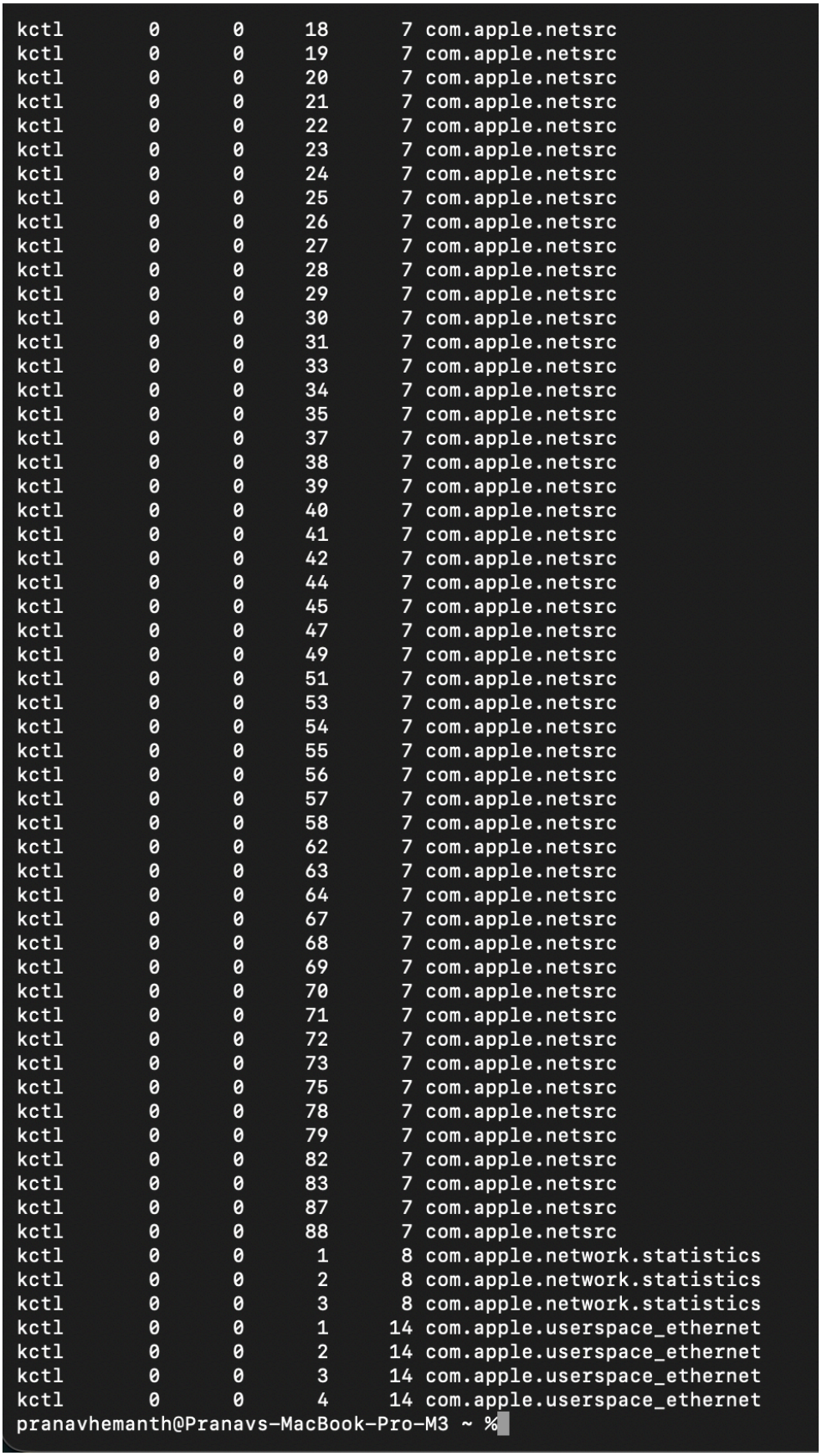
Command: ps aux



The ps aux command displays all active processes along with their resource usage and ownership, making it useful for spotting potentially suspicious activity.

30)Check Network Connections

Command: netstat -an



The netstat -an command shows active network connections and listening ports, helping to identify possible unauthorized communications.

31)Display Disk Information

Command: diskutil list

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %diskutil list
/dev/disk0 (internal, physical):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        *500.3 GB   disk0
   1:             Apple_APFS_ISC Container disk1         524.3 MB   disk0s1
   2:                 Apple_APFS Container disk3         494.4 GB   disk0s2
   3:        Apple_APFS_Recovery Container disk2         5.4 GB     disk0s3

/dev/disk3 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +494.4 GB   disk3
                                 Physical Store disk0s2
   1:                APFS Volume Macintosh HD            12.5 GB    disk3s1
   2:              APFS Snapshot com.apple.os.update-... 12.5 GB    disk3s1s1
   3:                APFS Volume Preboot                 8.9 GB     disk3s2
   4:                APFS Volume Recovery                1.3 GB     disk3s3
   5:                APFS Volume Data                    452.1 GB   disk3s5
   6:                APFS Volume VM                      9.7 GB     disk3s6

/dev/disk4 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        +934.8 MB   disk4
   1:                  Apple_HFS Claude                  934.8 MB   disk4s1

pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

This command displays comprehensive details about disk partitions and the storage layout on macOS systems.

32)Read macOS Version (System Artifact)

Command: cat /System/Library/CoreServices/SystemVersion.plist

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %cat /System/Library/CoreServices/SystemVersion.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
        <key>BuildID</key>
        <string>E0AF3C06-11FB-11F1-A7CC-608CDE06C496</string>
        <key>ProductBuildVersion</key>
        <string>25D2128</string>
        <key>ProductCopyright</key>
        <string>1983-2026 Apple Inc.</string>
        <key>ProductName</key>
        <string>macOS</string>
        <key>ProductUserVisibleVersion</key>
        <string>26.3.1</string>
        <key>ProductVersion</key>
        <string>26.3.1</string>
        <key>iOSSupportVersion</key>
        <string>26.3</string>
</dict>
</plist>
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %
```

This file stores the macOS version and build details, which can be used to identify the system during forensic analysis.

33)Read plist File in Human-Readable Format

Command: plutil -convert xml1 -o - ~/Library/Preferences/com.apple.finder.plist

```
                        </dict>
                </dict>
                <key>iconSize</key>
                <real>16</real>
                <key>scrollPositionX</key>
                <real>0.0</real>
                <key>scrollPositionY</key>
                <real>344</real>
                <key>showIconPreview</key>
                <true/>
                <key>sortColumn</key>
                <string>name</string>
                <key>textSize</key>
                <real>13</real>
                <key>useRelativeDates</key>
                <true/>
                <key>viewOptionsVersion</key>
                <integer>1</integer>
        </dict>
        <key>WindowState</key>
        <dict>
                <key>ContainerShowSidebar</key>
                <true/>
                <key>ShowSidebar</key>
                <true/>
                <key>ShowStatusBar</key>
                <false/>
                <key>ShowTabView</key>
                <false/>
                <key>ShowToolbar</key>
                <true/>
                <key>WindowBounds</key>
                <string>{{296, 398}, {920, 436}}</string>
        </dict>
/dict>
key>ViewSettingsDictionary</key>
dict>
        <key>Purple_Tag_ViewSettings</key>
        <dict>
                <key>WindowState</key>
                <dict>
                        <key>ContainerShowSidebar</key>
                        <true/>
                        <key>ShowSidebar</key>
                        <true/>
                        <key>ShowStatusBar</key>
                        <false/>
                        <key>ShowTabView</key>
                        <false/>
                        <key>ShowToolbar</key>
                        <true/>
                        <key>WindowBounds</key>
                        <string>{{2, 70}, {1228, 874}}</string>
                </dict>
        </dict>
/dict>


anth@Pranavs-MacBook-Pro-M3 ~ %
```

The plutil command transforms binary plist files into a readable XML format, making it easier to examine application and system preferences.

34)View Safari Browsing History

Command:

sqlite3 ~/Library/Safari/History.db

.tables

SELECT * FROM history_items LIMIT 20;

```
pranavhemanth@Pranavs-MacBook-Pro-M3 ~ %sqlite3 ~/Library/Safari/History.db
SQLite version 3.51.0 2025-06-12 13:14:41
Enter ".help" for usage hints.
sqlite> .tables
history_client_versions  history_items            history_tombstones
history_event_listeners  history_items_to_tags    history_visits
history_events           history_tags             metadata
sqlite> SELECT * FROM history_items LIMIT 20;
17|https://chatgpt.com/|chatgpt|316|^D^A|?^A|bplist00?^A\chatgpt.com/^H
|1|3320|0
25|https://meet.google.com/hqc-tuwy-pcd|meet.google|53|d||bplist00?^AUhqc-t^H
|0|200|0
29|https://meet.google.com/|meet.google|47|?||bplist00?^A_^P^Pmeet.google.com
|0|2710|0
42|https://meet.google.com/landing|meet.google|378|d|?^C|bplist00?^AVmeet g^H
|0|5500|0
52|https://www.youtube.com/|youtube|1054|d|?^D|bplist00?^A^B\youtube.com/_^P^
96|https://www.google.com/?client=safari|google|33|?|||0|275|400
121|https://accounts.google.com/gsi/transform|accounts.google|20|d|||0|600|20
125|https://www.canva.com/oauth/authorize/GOOGLE?origin=CFE&permissions=GET_B
0|300|200
263|https://www.google.com/search?q=g+meet&ie=UTF-8&oe=UTF-8&hl=en-in&client=
284|https://docs.google.com/document/u/0/?usp=direct_url|docs.google|57|d|?^A
285|https://docs.google.com/document/u/0/|docs.google|227|d|?|bplist00?^A^B^C
290|https://www.notion.com/|notion|10|,^A|?^A|bplist00?^A[notion.com/^H
|0|3500|0
301|https://www.notion.so/||2|,^A|?^A|bplist00?^AZnotion.so/^H
|0|3500|0
364|https://docs.google.com/document/d/1R3Mk0uFmgX8me29rRoTkjt9tngkF2btHFRrg-
365|https://docs.google.com/document/d/1R3Mk0uFmgX8me29rRoTkjt9tngkF2btHFRrg-
368|https://github.com/|github|744|d|0^B|bplist00?^A[github.com/^H
|1|1165|0
371|https://github.com/Pranavh-2004|github|632|^]^A|?^C||0|6410|0
372|https://github.com/Pranavh-2004?tab=repositories|github|403|^]^A|?^C|bpli
|0|6410|0
397|https://claude.ai/new||61|?|?||0|3600|0
399|https://github.com/new|github|146|d|?^A||0|3260|0
sqlite>
```

Safari keeps its browsing history in an SQLite database, which can be queried to retrieve user activity and visited URLs for forensic analysis.

Conclusion:

This lab covered fundamental digital forensics techniques such as filesystem analysis, memory acquisition, and artifact collection across both Linux and macOS systems. These approaches are crucial for examining system activity and detecting potential security incidents.